

Qualification Pack



Junior Cyber Security Associate

QP Code: NIE/ITS/Q0902

Version: 1.0

NSQF Level: 4

National Institute of Electronics and Information Technology (NIELIT) || NIELIT Bhawan, Plot No. 3, PSP
Pocket, Sector-8, Dwarka
New Delhi-110077 || email:puneetnema@nielit.gov.in

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NIE/ITS/Q0902: Junior Cyber Security Associate

Brief Job Description

A Junior Cyber Security Associate is an entry-level position that assists with protecting an organization's information systems and data from cyber threats. Common responsibilities include monitoring security systems, responding to security incidents, and conducting security audits. Qualifications typically include a bachelor's degree in a related field and strong technical skills. The job outlook is very good, with employment projected to grow much faster than average.

Personal Attributes

Junior Cyber Security Associates should be passionate about cybersecurity, have attention to detail and problem-solving skills, and be able to work under pressure and communicate effectively. They should also be curious, adaptable, and resilient.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [NIE/ITS/N0907: Operating System Basics-](#)
2. [NIE/ITS/N0908: Computer Network-](#)
3. [NIE/ITS/N0906: ARTIFICIAL INTELLIGENCE / MACHINE LEARNING AND DISRUPTIVE TECHNOLOGIES](#)
4. [NIE/ITS/N0905: PERCEPTION MANAGEMENT](#)
5. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Information Technology Sector
Sub-Sector	Networking and Cybersecurity
Occupation	Cyber Security
Country	India
NSQF Level	4
Credits	15

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Aligned to NCO/ISCO/ISIC Code	2522.0201 Security Analyst; 3513.0200 Computer Security Specialist
Minimum Educational Qualification & Experience	12th grade Pass OR Completed 2nd year of the 3-year diploma after 10 (and pursuing regular diploma in CS/IT/EC/EE/ allied branches) OR 10th grade pass (plus 2-year NTC in IT-ITeS sector) OR 10th grade pass with 1 year NTC plus NAC (in IT-ITeS sector) OR 10th grade pass and pursuing continuous schooling OR Previous relevant Qualification of NSQF Level (Level-3* with 2 Years of experience in IT-ITeS Sector)
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	16 Years
Last Reviewed On	NA
Next Review Date	30/06/2026
NSQC Approval Date	29/03/2023
Version	1.0
Reference code on NQR	QG-04-IT-00351-2023-V1-NIELIT
NQR Version	1

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NIE/ITS/N0907: Operating system Basics

Description

Fundamentals of Network and Operating Systems encompass competencies in understanding networking fundamentals, operating system basics, network configuration and management, network services and protocols, troubleshooting, virtualization, security, documentation, and compliance within the context of network and operating system environments.

Scope

The scope covers the following :

- Networking fundamentals, configuration and management, operating system basics, virtualization, security

Elements and Performance Criteria

Computer Networking

To be competent, the user/individual on the job must be able to:

- PC1.** Knowledge about the Benefits of Networks, components of Computer Networks and also have an insight into various topologies used in networking.
- PC2.** Explain various Transmission modes, Ethernet, and the Classification Of Transmission Media.
- PC3.** In-depth idea about various networking devices like Repeaters, Hubs, Bridges, Switches, and routers

OSI Reference Model & TCP/IP Protocol Suite

To be competent, the user/individual on the job must be able to:

- PC4.** Overview of OSI, TCP/IP Model, and Function of layers, learn about various types of networks.
- PC5.** Knowledge about TCP/IP Protocol Model in Depth, Port Numbers, Common TCP& UDP, Ports, and also Some important TCP/IP protocols
- PC6.** About IP addressing, classes, IP addressing scheme, subnetting basics, how to create subnets, subnet masks, subnetting Class C addresses, subnetting Class B addresses, and VLSM include: efficiency, scalability, flexibility, and security.

TCP/IP command line Troubleshooting utilities

To be competent, the user/individual on the job must be able to:

- PC7.**
 - Configure ip addresses in Windows and Linux and also have a proper introduction to RIR, registries and IANA along with attaining knowledge about TCP/IP
 - Troubleshooting utilities
- PC8.** Troubleshooting IP Addressing

Overview and Exploring Windows OS

To be competent, the user/individual on the job must be able to:

- PC9.** In-depth idea about the Windows operating system and its File System
- PC10.** Understand Basic Windows and its Control Panel
- PC11.** Understand various management service roles and get acquainted with the concept of remote desktop management.

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PC12. Students will be able to install the Windows Operating system

Overview and Exploring Linux OS

To be competent, the user/individual on the job must be able to:

PC13. The in-depth idea about the Linux operating system and its File System

PC14. Working on Command Line Utilities, Managing Users and Groups, Configuring Permissions, Read, Write, and Execute

PC15. Preparing the File System for ACLs, Configuring Administrator Tasks with sudo

PC16. Working with Links: Working with Symbolic Links, Working with Hard Links

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Computer Networking: Understanding the fundamentals of computer networking, including how data is transmitted, network topologies, common network protocols, and the importance of networks in modern computing.

KU2. OSI Reference Model & TCP/IP Protocol Suite: Mastery of the OSI reference model, which breaks down network communication into seven layers, and the TCP/IP protocol suite, which is used in the internet and real-world networking scenarios, allowing for a deep understanding of how data is encapsulated and transmitted across networks.

KU3. TCP/IP Command Line Troubleshooting Utilities: Proficiency in using command-line tools such as ping, traceroute, and ipconfig to diagnose and troubleshoot network-related issues, enhancing the ability to resolve connectivity and communication problems effectively.

KU4. Overview and Exploring Windows OS: Familiarity with the Windows operating system, including its interface, file system, user management, and basic administrative tasks, enabling students to navigate and operate within a Windows environment.

KU5. Overview and Exploring Linux OS: Competence in the Linux operating system, understanding its structure, file system, user and group management, and basic system administration tasks, allowing students to effectively use and administer Linux-based systems.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. A strong foundation in computer networking, OSI and TCP/IP models, command-line troubleshooting, and proficiency in Windows and Linux operating systems, providing a comprehensive understanding of core IT concepts.

GS2. A valuable set of general skills, including critical thinking, effective communication, time management, adaptability, and attention to detail, making them well-rounded professionals in the IT field.

GS3. Professional skills in network configuration, protocol understanding, operating system proficiency, security awareness, problem-solving, documentation, and user and group management, equipping them for roles in network administration, system administration, IT support, and providing a platform for continuous learning and career growth in the technology industry.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Computer Networking</i>	-	-	-	-
PC1. Knowledge about the Benefits of Networks, components of Computer Networks and also have an insight into various topologies used in networking.	-	-	-	-
PC2. Explain various Transmission modes, Ethernet, and the Classification Of Transmission Media.	-	-	-	-
PC3. In-depth idea about various networking devices like Repeaters, Hubs, Bridges, Switches, and routers	-	-	-	-
<i>OSI Reference Model & TCP/IP Protocol Suite</i>	-	-	-	-
PC4. Overview of OSI, TCP/IP Model, and Function of layers, learn about various types of networks.	-	-	-	-
PC5. Knowledge about TCP/IP Protocol Model in Depth, Port Numbers, Common TCP& UDP, Ports, and also Some important TCP/IP protocols	-	-	-	-
PC6. About IP addressing, classes, IP addressing scheme, subnetting basics, how to create subnets, subnet masks, subnetting Class C addresses, subnetting Class B addresses, and VLSM include: efficiency, scalability, flexibility, and security.	-	-	-	-
<i>TCP/IP command line Troubleshooting utilities</i>	-	-	-	-
PC7. • Configure ip addresses in Windows and Linux and also have a proper introduction to RIR, registries and IANA along with attaining knowledge about TCP/IP • Troubleshooting utilities	-	-	-	-
PC8. Troubleshooting IP Addressing	-	-	-	-
<i>Overview and Exploring Windows OS</i>	-	-	-	-
PC9. In-depth idea about the Windows operating system and its File System	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Understand Basic Windows and its Control Panel	-	-	-	-
PC11. Understand various management service roles and get acquainted with the concept of remote desktop management.	-	-	-	-
PC12. Students will be able to install the Windows Operating system	-	-	-	-
<i>Overview and Exploring Linux OS</i>	-	-	-	-
PC13. The in-depth idea about the Linux operating system and its File System	-	-	-	-
PC14. Working on Command Line Utilities, Managing Users and Groups, Configuring Permissions, Read, Write, and Execute	-	-	-	-
PC15. Preparing the File System for ACLs, Configuring Administrator Tasks with sudo	-	-	-	-
PC16. Working with Links: Working with Symbolic Links, Working with Hard Links	-	-	-	-
NOS Total	9	4	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0907
NOS Name	Operating system Basics
Sector	Information Technology Sector
Sub-Sector	Networking and Cybersecurity
Occupation	Cyber Security
NSQF Level	5
Credits	3.5
Version	1.0
Next Review Date	29/03/2026

Qualification Pack

NIE/ITS/N0907: Operating System Basics-

Description

Fundamentals of Network and Operating Systems encompass competencies in understanding networking fundamentals, operating system basics, network configuration and management, network services and protocols, troubleshooting, virtualization, security, documentation, and compliance within the context of network and operating system environments.

Scope

The scope covers the following :

- Networking fundamentals, configuration and management, operating system basics, virtualization, security

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Overview of OSI, TCP/IP Model and Function of layers, learn about various types of networks.
- PC2.** Knowledge about the Benefits of Networks, and components of Computer Networks and also have an insight into various topologies used in networking.
- PC3.** Explain various Transmission modes, Ethernet, and the Classification Of Transmission Media.
- PC4.** In-depth idea about various networking devices like Repeaters, Hubs, Bridges, Switches, and routers
- PC5.** Introduction to TCP/IP Protocol Model in Depth, Port Numbers, Common TCP& UDP Ports and also Some important TCP/IP protocols.
- PC6.** IP addressing, Classes, IP Addressing Scheme, Subnetting Basics alongwith How to Create Subnets, Subnet Masks, Subnetting Class C Addresses, Subnetting Class B Addresses and VLSM
- PC7.** Configure ip addresses in Windows and Linux and also have a proper introduction to RIR, registries and IANA along with attaining knowledge about TCP/IP Troubleshooting utilities, Troubleshooting IP Addressing
- PC8.** The in-depth idea about the Windows and Linux operating system and its File System
- PC9.** . Working on Command Line Utilities, Managing Users and Groups, Configuring Permissions, Read, Write, and Execute, Preparing the File System for ACLs
- PC10.** Working with Links: Working with Symbolic Links, Working with Hard Links

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Computer Networking: Understanding the fundamentals of computer networking, including how data is transmitted, network topologies, common network protocols, and the importance of networks in modern computing.

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- KU2.** OSI Reference Model & TCP/IP Protocol Suite: Mastery of the OSI reference model, which breaks down network communication into seven layers, and the TCP/IP protocol suite, which is used in the internet and real-world networking scenarios, allowing for a deep understanding of how data is encapsulated and transmitted across networks.
- KU3.** TCP/IP Command Line Troubleshooting Utilities: Proficiency in using command-line tools such as ping, tracert, and ipconfig to diagnose and troubleshoot network-related issues, enhancing the ability to resolve connectivity and communication problems effectively.
- KU4.** Overview and Exploring Windows OS: Familiarity with the Windows operating system, including its interface, file system, user management, and basic administrative tasks, enabling students to navigate and operate within a Windows environment.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** A strong foundation in computer networking, OSI and TCP/IP models, command-line troubleshooting, and proficiency in Windows and Linux operating systems, providing a comprehensive understanding of core IT concepts.
- GS2.** A valuable set of general skills, including critical thinking, effective communication, time management, adaptability, and attention to detail, making them well-rounded professionals in the IT field.
- GS3.** Professional skills in network configuration, protocol understanding, operating system proficiency, security awareness, problem-solving, documentation, and user and group management, equipping them for roles in network administration, system administration, IT support, and providing a platform for continuous learning and career growth in the technology industry.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	9	4	-	-
PC1. Overview of OSI, TCP/IP Model and Function of layers, learn about various types of networks.	-	-	-	-
PC2. Knowledge about the Benefits of Networks, and components of Computer Networks and also have an insight into various topologies used in networking.	-	-	-	-
PC3. Explain various Transmission modes, Ethernet, and the Classification Of Transmission Media.	-	-	-	-
PC4. In-depth idea about various networking devices like Repeaters, Hubs, Bridges, Switches, and routers	-	-	-	-
PC5. Introduction to TCP/IP Protocol Model in Depth, Port Numbers, Common TCP& UDP Ports and also Some important TCP/IP protocols.	-	-	-	-
PC6. IP addressing, Classes, IP Addressing Scheme, Subnetting Basics alongwith How to Create Subnets, Subnet Masks, Subnetting Class C Addresses, Subnetting Class B Addresses and VLSM	-	-	-	-
PC7. Configure ip addresses in Windows and Linux and also have a proper introduction to RIR, registries and IANA along with attaining knowledge about TCP/IP Troubleshooting utilities, Troubleshooting IP Addressing	-	-	-	-
PC8. The in-depth idea about the Windows and Linux operating system and its File System	-	-	-	-
PC9. . Working on Command Line Utilities, Managing Users and Groups, Configuring Permissions, Read, Write, and Execute, Preparing the File System for ACLs	-	-	-	-
PC10. Working with Links: Working with Symbolic Links, Working with Hard Links	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	9	4	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0907
NOS Name	Operating System Basics-
Sector	Information Technology Sector
Sub-Sector	Networking and Cybersecurity
Occupation	Cyber Security
NSQF Level	5
Credits	0.7
Version	1.0
Next Review Date	NA

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NIE/ITS/N0908: Computer Network*

Description

Fundamentals of Cyber Security encompass competencies in core cyber security principles, regulations and policies, threat awareness, network and operating system security, access control, incident response, data protection, ethical hacking, and risk management, providing a framework for individuals to understand, implement, and adhere to essential security practices in the field of cyber security.

Scope

The scope covers the following :

- Fundamentals of Cyber Security encompasses the foundational knowledge and skills required to comprehend, implement, and maintain essential security practices, including threat awareness, network and system security, access control, and incident response, to safeguard digital assets and data from cyber threats and vulnerabilities.

Elements and Performance Criteria

Basics of Information Security

To be competent, the user/individual on the job must be able to:

- PC1.** understand Information Security, CIA Triad, Confidentiality, integrity, Availability Authentication, Authorization, and Accounting.
- PC2.** Understanding computer Ethics will be able to the differentiate between ethical and Unethical behavior
- PC3.** About Services & Techniques, Actives, Passive Threats and Exploit, etc, Authentication Authorization and Accounting (AAA)

Ethical Hacking and Basic Cryptography

To be competent, the user/individual on the job must be able to:

- PC4.** Overview of Ethical Hacking.
- PC5.** About the types of malware and its prevention.
- PC6.** About Cryptography, Plaintext, Ciphertext, Encryption, Decryption, Cipher, Symmetric, and asymmetric Cryptography, Hashing

Introduction to OS security

To be competent, the user/individual on the job must be able to:

- PC7.** About OS security and Access Control.
- PC8.** Share Security, Password Policies, User Account Policies.

Security Tools for Windows Operating System

To be competent, the user/individual on the job must be able to:

- PC9.** Overview of security tools for Windows Operating System.
- PC10.** About Virus & Threat Protection mechanisms.
- PC11.** About Firewall & Network Protection, install and configure a personal firewall, and will be able to create Inbound & Outbound Rules

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Introduction to Wireless Network and Wireless Security

To be competent, the user/individual on the job must be able to:

PC12. About Wireless Networks and their Security

Android and iPhone iOS Operating systems

To be competent, the user/individual on the job must be able to:

PC13. About the security risks involved in Android OS and iOS.

Web Application Protocols & Browser Security

To be competent, the user/individual on the job must be able to:

PC14. About Web application terminologies and Browser Security.

Introduction to Social Media & its security

To be competent, the user/individual on the job must be able to:

PC15. About Social Media platforms and various cybercrimes

Digital Payment System and Security

To be competent, the user/individual on the job must be able to:

PC16. About the Cashless Economy and Payments Terminology,

PC17. Types of Payment Processing, and Security Concerns in Digital Payment System.

PC18. Various kinds of fraud in banking

Overview on Cyber Law & Cyber Cell

To be competent, the user/individual on the job must be able to:

PC19. About Cyber Crime and the background of Cyber Law.

PC20. About Cyber Cell & Cyber Fraud and helpline systems.

PC21. About registering an Online Cyber Crime, Complaint, and Registration of cybercrime complaints, Cyber Fraud Helpline

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** • Basics of Information Security: The importance of information security, CIA triad (confidentiality, integrity, and availability)
• Common security threats and vulnerabilities, best practices for protecting information
- KU2.** Ethical Hacking and Basic Cryptography: The principles and techniques of ethical hacking, How to identify and exploit security vulnerabilities, Basic cryptography concepts, such as encryption and decryption
- KU3.** Introduction to OS Security: The security features of popular operating systems, such as Windows, macOS, and Linux, How to harden operating systems to improve security, Common operating system security threats and vulnerabilities
- KU4.** Security Tools for Windows Operating System: How to use popular security tools to protect Windows systems, How to monitor and manage Windows security, Common Windows security vulnerabilities and how to mitigate them
- KU5.** Introduction to Wireless Network and Wireless Security: The basics of wireless networking, Common wireless security threats and vulnerabilities, Best practices for securing wireless networks

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- KU6.** Introduction to the Android and iPhone iOS Operating systems: The security features of Android and iOS devices, How to harden Android and iOS devices to improve security, Common Android and iOS security threats and vulnerabilities, Android and iPhone iOS Operating systems, A more in-depth study of the security features of Android and iOS devices, How to develop secure Android and iOS apps, Advanced Android and iOS security threats and vulnerabilities
- KU7.** Web Application & Browser Security: The basics of web applications, Common web application security threats and vulnerabilities, Best practices for securing web applications, Web Application Protocols & Browser Security, The protocols used by web applications, Common web application protocol vulnerabilities, How to secure web browsers
- KU8.** Introduction to Social Media & its security: The basics of social media, Common social media security threats and vulnerabilities, Best practices for securing social media accounts
- KU9.** Digital Payment System and Security: The basics of digital payment systems, Common digital payment system security threats and vulnerabilities, Best practices for securing digital payments, Various kinds of fraud in banking, Common types of banking fraud, How to detect and prevent banking fraud
- KU10.** Overview on Cyber Law & Cyber Cell: The laws related to cyber crime in India, The role of the cyber cell in investigating and prosecuting cyber crime, Cyber Fraud Helpline, How to report cyber fraud to the authorities, What to do if you are a victim of cyber fraud

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Students studying these topics will develop general skills such as critical thinking, problem-solving, and effective communication to assess and address various cybersecurity challenges.
- GS2.** Students will gain professional skills, including the ability to apply ethical hacking techniques, secure operating systems and mobile platforms, manage digital payment systems securely, and understand the legal and law enforcement aspects of cybercrime and fraud prevention.
- GS3.** These combined general and professional skills will prepare students for careers in cybersecurity, digital forensics, or related fields, where they can safeguard digital assets, investigate cyber incidents, and contribute to the overall security of information systems.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Basics of Information Security</i>	-	-	-	-
PC1. understand Information Security, CIA Triad, Confidentiality, integrity, Availability Authentication, Authorization, and Accounting.	-	-	-	-
PC2. Understanding computer Ethics will be able to the differentiate between ethical and Unethical behavior	-	-	-	-
PC3. About Services & Techniques, Actives, Passive Threats and Exploit, etc, Authentication Authorization and Accounting (AAA)	-	-	-	-
<i>Ethical Hacking and Basic Cryptography</i>	-	-	-	-
PC4. Overview of Ethical Hacking.	-	-	-	-
PC5. About the types of malware and its prevention.	-	-	-	-
PC6. About Cryptography, Plaintext, Ciphertext, Encryption, Decryption, Cipher, Symmetric, and asymmetric Cryptography, Hashing	-	-	-	-
<i>Introduction to OS security</i>	-	-	-	-
PC7. About OS security and Access Control.	-	-	-	-
PC8. Share Security, Password Policies, User Account Policies.	-	-	-	-
<i>Security Tools for Windows Operating System</i>	-	-	-	-
PC9. Overview of security tools for Windows Operating System.	-	-	-	-
PC10. About Virus & Threat Protection mechanisms.	-	-	-	-
PC11. About Firewall & Network Protection, install and configure a personal firewall, and will be able to create Inbound & Outbound Rules	-	-	-	-
<i>Introduction to Wireless Network and Wireless Security</i>	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. About Wireless Networks and their Security	-	-	-	-
<i>Android and iPhone iOS Operating systems</i>	-	-	-	-
PC13. About the security risks involved in Android OS and iOS.	-	-	-	-
<i>Web Application Protocols & Browser Security</i>	-	-	-	-
PC14. About Web application terminologies and Browser Security.	-	-	-	-
<i>Introduction to Social Media & its security</i>	-	-	-	-
PC15. About Social Media platforms and various cybercrimes	-	-	-	-
<i>Digital Payment System and Security</i>	-	-	-	-
PC16. About the Cashless Economy and Payments Terminology,	-	-	-	-
PC17. Types of Payment Processing, and Security Concerns in Digital Payment System.	-	-	-	-
PC18. Various kinds of fraud in banking	-	-	-	-
<i>Overview on Cyber Law & Cyber Cell</i>	-	-	-	-
PC19. About Cyber Crime and the background of Cyber Law.	-	-	-	-
PC20. About Cyber Cell & Cyber Fraud and helpline systems.	-	-	-	-
PC21. About registering an Online Cyber Crime, Complaint, and Registration of cybercrime complaints, Cyber Fraud Helpline	-	-	-	-
NOS Total	55	27	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0908
NOS Name	Computer Network*
Sector	Information Technology Sector
Sub-Sector	Networking and Cybersecurity
Occupation	Cyber Security
NSQF Level	5
Credits	4.1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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NIE/ITS/N0908: Computer Network-

Description

develops the competencies required to design, configure, and manage secure computer networks. It covers foundational and advanced networking concepts including IP addressing (IPv4/IPv6), subnetting, routing protocols (RIP, OSPF, EIGRP), VLANs, ACLs, NAT, and WAN technologies. Learners gain hands-on experience with configuring network devices, securing Layer 2 environments, enabling remote access protocols, and implementing monitoring and VPN solutions.

Scope

The scope covers the following :

- It aligns with the National Occupational Standards (NOS) by preparing learners for roles in network administration, technical support, and cybersecurity. It covers practical skills such as configuring routers, switches, protocols, and network security features, making students job-ready for roles in IT infrastructure. The hands-on focus on troubleshooting, monitoring, and securing networks supports core competencies expected in national and global ICT job markets.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Learner can successfully design and configure LAN/WAN networks using IPv4/IPv6 addressing, subnetting (FLSM/VLSM), VLANs, and routing protocols (RIP, OSPF, EIGRP).
- PC2.** Demonstrates the ability to configure network devices (routers/switches) using CLI for services like SSH, Telnet, NAT, ACLs, and VPNs, including advanced protocols like HSRP, VRRP, and GLBP.
- PC3.** Accurately troubleshoots and verifies configuration and performance of routing, switching, NAT, WAN protocols (PPP, HDLC, Frame Relay), and network connectivity using appropriate tools and logs.
- PC4.** Effectively implements network security measures such as port-security, ACLs, and VPN tunnels, and configures monitoring/logging tools like SNMP and syslog servers to track device activities.
- PC5.** Configures and validates NTP server-client setups, interprets routing tables, spanning-tree protocol status, and uses discovery protocols (CDP, LLDP) to assess and document network topology.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Develop a solid understanding of network types, topologies, and communication models (OSI and TCP/IP), including the roles of various protocols across layers.
- KU2.** Gain practical knowledge in IP addressing (IPv4/IPv6), subnetting (FLSM/VLSM), and configuring network devices using command-line interfaces.

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- KU3.** Learn to implement and troubleshoot routing protocols (RIP, OSPF, EIGRP), access control mechanisms, and Layer 2 security features such as VLANs and port-security.
- KU4.** Acquire skills in configuring WAN technologies, VPNs, and essential network services like NTP, SNMP, and logging for secure and synchronized network operations.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Proficient in designing, configuring, and managing various network topologies, IP addressing (IPv4/IPv6), and routing protocols including static, dynamic (RIP, OSPF, EIGRP), and advanced routing techniques.
- GS2.** Skilled in configuring and securing network devices such as routers and switches, including VLANs, trunking, ACLs, NAT, and implementing Layer 2 and Layer 3 security features like port-security and VPNs.
- GS3.** Experienced in troubleshooting network connectivity, routing issues, WAN protocols (PPP, HDLC, Frame Relay), and ensuring network performance through monitoring tools, SNMP, and logging.
- GS4.** Knowledgeable in network time synchronization (NTP), network traffic control, and applying security best practices to safeguard data transmission across LAN, WAN, and internet environments.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	55	27	-	-
PC1. Learner can successfully design and configure LAN/WAN networks using IPv4/IPv6 addressing, subnetting (FLSM/VLSM), VLANs, and routing protocols (RIP, OSPF, EIGRP).	55	27	-	-
PC2. Demonstrates the ability to configure network devices (routers/switches) using CLI for services like SSH, Telnet, NAT, ACLs, and VPNs, including advanced protocols like HSRP, VRRP, and GLBP.	-	-	-	-
PC3. Accurately troubleshoots and verifies configuration and performance of routing, switching, NAT, WAN protocols (PPP, HDLC, Frame Relay), and network connectivity using appropriate tools and logs.	-	-	-	-
PC4. Effectively implements network security measures such as port-security, ACLs, and VPN tunnels, and configures monitoring/logging tools like SNMP and syslog servers to track device activities.	-	-	-	-
PC5. Configures and validates NTP server-client setups, interprets routing tables, spanning-tree protocol status, and uses discovery protocols (CDP, LLDP) to assess and document network topology.	-	-	-	-
NOS Total	55	27	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0908
NOS Name	Computer Network-
Sector	Information Technology Sector
Sub-Sector	Networking and Cybersecurity
Occupation	Cyber Security
NSQF Level	5
Credits	4.1
Version	1.0
Next Review Date	NA

Qualification Pack

NIE/ITS/N0906: Cryptography and Ethical Hacking

Description

Cryptography and Ethical Hacking encompass competencies in cryptographic principles, ethical hacking practices, information security, cyber tools, network security, legal and ethical considerations, incident response, digital forensics, reporting, and a commitment to continuous learning, equipping professionals with the skills and knowledge needed for careers in cybersecurity, penetration testing, and digital forensics.

Scope

The scope covers the following :

- Cryptography and Ethical Hacking encompass a comprehensive range of competencies in information security, cryptographic techniques, ethical hacking, security tools, and legal and ethical considerations, preparing individuals to secure digital assets and perform ethical assessments of information systems and networks while adhering to professional standards.

Elements and Performance Criteria

Introduction to Cryptography and Public Key Infrastructure (PKI)

To be competent, the user/individual on the job must be able to:

- PC1.** Overview of Cryptography and Classical and Modern Cryptography.
- PC2.** Overview of public Key Infrastructure (PKI).

Scanning and Information Gathering

To be competent, the user/individual on the job must be able to:

- PC3.** Understand Scanning and Sniffing and use the NMAP and Wireshark tools for scanning and sniffing.
- PC4.** About Detecting Network Attacks using Wireshark.
- PC5.** Overview of Information Gathering.

Open-source intelligence (OSINT) and Man in the Middle (MITM) Attacks

To be competent, the user/individual on the job must be able to:

- PC6.** Overview of Open-source intelligence (OSINT) framework.
- PC7.** Overview of MITM Attacks and their Countermeasures.

Password Cracking and Countermeasures

To be competent, the user/individual on the job must be able to:

- PC8.** Understand the types of password attacks.
- PC9.** Understand the Hash function, Hash algorithm, Password Hashes, and Password Cracking Countermeasures.
- PC10.** Understand the various types of Spoofing, Denial of Service attacks, and their countermeasures.

Steganography and E-Mail Security

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC11.** Understand the Steganography and Information Hiding Techniques.
- PC12.** Concept of Email and their protocols and understanding Email Spoofing and Phishing Techniques.
- PC13.** Understanding the concepts of Fake E-mails.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Students will acquire a deep understanding of cryptography principles, encryption algorithms, and the implementation of secure communication methods, enabling them to protect data at rest and in transit effectively.
- KU2.** They will develop expertise in ethical hacking, penetration testing, and security auditing techniques, allowing them to identify vulnerabilities, assess security postures, and contribute to the proactive defense of information systems.
- KU3.** Students will gain knowledge of legal and ethical frameworks, incident response and digital forensics, network and web application security, as well as compliance requirements, equipping them with a comprehensive grasp of the ethical and technical aspects of cybersecurity and ethical hacking practices.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Students who engage with Cryptography and Ethical Hacking acquire general skills such as critical thinking, problem-solving, and ethical decision-making, enabling them to analyze and address complex security challenges effectively.
- GS2.** Professionally, students will develop technical skills in cryptographic techniques, ethical hacking methodologies, and vulnerability assessment, preparing them for roles in cybersecurity, digital forensics, and security auditing, while ensuring they adhere to ethical and legal standards in their practice.
- GS3.** Students cultivate communication skills that allow them to collaborate effectively with teams, communicate security findings, and contribute to the development of comprehensive security policies, enhancing their overall professional readiness in the field of information security.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Cryptography and Public Key Infrastructure (PKI)</i>	10	3	-	-
PC1. Overview of Cryptography and Classical and Modern Cryptography.	-	-	-	-
PC2. Overview of public Key Infrastructure (PKI).	-	-	-	-
<i>Scanning and Information Gathering</i>	15	3	1	-
PC3. Understand Scanning and Sniffing and use the NMAP and Wireshark tools for scanning and sniffing.	-	-	-	-
PC4. About Detecting Network Attacks using Wireshark.	-	-	-	-
PC5. Overview of Information Gathering.	-	-	-	-
<i>Open-source intelligence (OSINT) and Man in the Middle (MITM) Attacks</i>	15	3	1	-
PC6. Overview of Open-source intelligence (OSINT) framework.	-	-	-	-
PC7. Overview of MITM Attacks and their Countermeasures.	-	-	-	-
<i>Password Cracking and Countermeasures</i>	15	4	1	-
PC8. Understand the types of password attacks.	-	-	-	-
PC9. Understand the Hash function, Hash algorithm, Password Hashes, and Password Cracking Countermeasures.	-	-	-	-
PC10. Understand the various types of Spoofing, Denial of Service attacks, and their countermeasures.	-	-	-	-
<i>Steganography and E-Mail Security</i>	11	3	1	-
PC11. Understand the Steganography and Information Hiding Techniques.	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Concept of Email and their protocols and understanding Email Spoofing and Phishing Techniques.	-	-	-	-
PC13. Understanding the concepts of Fake E-mails.	-	-	-	-
NOS Total	66	16	4	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0906
NOS Name	Cryptography and Ethical Hacking
Sector	Information Technology Sector
Sub-Sector	Networking and Cybersecurity
Occupation	Cyber Security
NSQF Level	4
Credits	2.5
Version	1.0
Next Review Date	29/03/2026

Qualification Pack

NIE/ITS/N0906: ARTIFICIAL INTELLIGENCE / MACHINE LEARNING AND DISRUPTIVE TECHNOLOGIES

Description

This module equips learners with foundational and applied skills in Artificial Intelligence (AI) and Machine Learning (ML) using Python, with a focus on social media applications. It also covers essential system administration tasks for Windows and Linux servers, including operating system-level security features. Additionally, learners are introduced to disruptive technologies such as Big Data and Cloud Computing, enabling them to understand and navigate modern digital infrastructures effectively.

Scope

The scope covers the following :

- It equips learners with hands-on skills in managing and securing both Windows and Linux servers, essential for modern IT infrastructure. It introduces AI/ML concepts using Python with a focus on applications in social media analytics. Additionally, it offers foundational knowledge of disruptive technologies like Big Data and Cloud Computing, preparing students for roles in data-driven and cloud-based environments. Overall, it bridges system administration with emerging tech trends and intelligent automation.

Elements and Performance Criteria

AI/ML using Python in Social Media Context

To be competent, the user/individual on the job must be able to:

- PC1.** Apply Python libraries (e.g., Scikit-learn, Pandas, TensorFlow) to build basic AI/ML models.
- PC2.** Analyze social media datasets to detect trends, sentiments, and predictive patterns.
- PC3.** Integrate AI models into social media platforms for real-time decision-making or recommendation systems.

Windows Server Administration

To be competent, the user/individual on the job must be able to:

- PC4.** Configure and manage core services (Active Directory, DNS, DHCP) in a Windows Server environment.
- PC5.** Implement user and group policies to enhance system security and manage access rights.
- PC6.** Monitor and troubleshoot server performance and security using built-in tools.

Configure and secure Linux-based server systems.

To be competent, the user/individual on the job must be able to:

- PC7.** Install and configure services (e.g., SSH, Apache, firewall settings) on Linux.
- PC8.** Implement user permissions and access controls using Linux security models.
- PC9.** Apply OS-level security measures (e.g., SELinux, audit logs, iptables).

Introduction to Disruptive Technologies (Big Data & Cloud Computing)

To be competent, the user/individual on the job must be able to:

- PC10.** Describe key concepts and use cases of Big Data and Cloud Computing.

Qualification Pack

PC11. Identify major platforms (e.g., Hadoop, AWS, Azure) and their services.

PC12. Evaluate the benefits and challenges of adopting disruptive technologies in industry.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Learn to build machine learning models using Python and apply them to analyze social media data, such as sentiment and trends.
- KU2.** Understand how to install, configure, and manage Windows and Linux servers, including user roles and network services.
- KU3.** Learn to secure servers using built-in OS tools like firewalls, SELinux, and update management to prevent common threats.
- KU4.** Get introduced to cloud platforms and Big Data tools, exploring how they support large-scale data processing and modern IT services.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Proficient in Python-based AI/ML development and social media data analysis using popular libraries and natural language processing techniques.
- GS2.** Skilled in installing, configuring, and securing both Windows and Linux server environments, including user management, permissions, and system monitoring.
- GS3.** Experienced in applying OS-level security features such as firewalls, patch management, SELinux/AppArmor, and secure access protocols.
- GS4.** Knowledgeable about disruptive technologies, including Big Data frameworks and cloud computing models, with the ability to leverage these for scalable and innovative IT solutions.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>AI/ML using Python in Social Media Context</i>	17	9	-	-
PC1. Apply Python libraries (e.g., Scikit-learn, Pandas, TensorFlow) to build basic AI/ML models.	-	-	-	-
PC2. Analyze social media datasets to detect trends, sentiments, and predictive patterns.	-	-	-	-
PC3. Integrate AI models into social media platforms for real-time decision-making or recommendation systems.	-	-	-	-
<i>Windows Server Administration</i>	-	-	-	-
PC4. Configure and manage core services (Active Directory, DNS, DHCP) in a Windows Server environment.	-	-	-	-
PC5. Implement user and group policies to enhance system security and manage access rights.	-	-	-	-
PC6. Monitor and troubleshoot server performance and security using built-in tools.	-	-	-	-
<i>Configure and secure Linux-based server systems.</i>	-	-	-	-
PC7. Install and configure services (e.g., SSH, Apache, firewall settings) on Linux.	-	-	-	-
PC8. Implement user permissions and access controls using Linux security models.	-	-	-	-
PC9. Apply OS-level security measures (e.g., SELinux, audit logs, iptables).	-	-	-	-
<i>Introduction to Disruptive Technologies (Big Data & Cloud Computing)</i>	-	-	-	-
PC10. Describe key concepts and use cases of Big Data and Cloud Computing.	-	-	-	-
PC11. Identify major platforms (e.g., Hadoop, AWS, Azure) and their services.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Evaluate the benefits and challenges of adopting disruptive technologies in industry.	-	-	-	-
NOS Total	17	9	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0906
NOS Name	ARTIFICIAL INTELLIGENCE / MACHINE LEARNING AND DISRUPTIVE TECHNOLOGIES
Sector	Information Technology Sector
Sub-Sector	
Occupation	CYBER SECURITY
NSQF Level	5
Credits	1.3
Version	1.0
Next Review Date	NA

Qualification Pack

NIE/ITS/N0905: Network and Infrastructure Security

Description

Network and Infrastructure Security encompass competencies in core security principles, including network and infrastructure protection, access control, incident response, encryption, risk management, and compliance, enabling individuals to effectively secure and manage digital assets and networks in various professional contexts.

Scope

The scope covers the following :

- Network and Infrastructure Security encompass competencies related to securing network and infrastructure components, including network security, operating system security, access control, incident response, vulnerability management, and adherence to security policies and compliance, providing a framework for individuals to understand, implement, and maintain essential security practices in the context of network and infrastructure protection.

Elements and Performance Criteria

Introduction to Network Security and L2/L3 Attacks

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the concept of Network Security and Network Security Protocols.
- PC2.** Understand the Attacks at Layer 3 and Layer 2
- PC3.** Understand the CAM Table and CAM Flooding Attacks.
- PC4.** Understand the Switch Port Security and Switch Port Violations.
- PC5.** Understand the concept to prevent CAM Flooding Attacks by using Switch Port Security

Network Security Control

To be competent, the user/individual on the job must be able to:

- PC6.** Understand the concept of Network Security Controls and Network Access controls
- PC7.** Understand the concept of Firewalls, DMZ, VPN, IDS, IPS, and Network Traffic Monitoring and Analysis.
- PC8.** Understand the Working of VPN, VPN Components and VPN Types.

Intrusion Detection System(IDS)

To be competent, the user/individual on the job must be able to:

- PC9.** Understand the working of IP Security, Protocols used in IPSec, the Security Architecture of IPSec, and Modes of IPSec.
- PC10.** Understand the concept of Intrusion Detection.

Understanding SOC

To be competent, the user/individual on the job must be able to:

- PC11.** Understand SOC, NOC, and the Functions of SOC.
- PC12.** Understand SIEM and Security Operations and Management

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Learners will gain knowledge of fundamental network security principles, including protocols, attack types at different OSI layers, CAM table and CAM flooding attacks, and configuring switch port security.
- KU2.** Learners will understand how to prevent CAM flooding attacks, implement network access controls, and work with security measures like firewalls, VPNs, IDS/IPS, and traffic analysis.
- KU3.** Learners will grasp IP security, intrusion detection, the functions of Security Operations Centers (SOCs) and Network Operations Centers (NOCs), and the role of SIEM systems in security operations and management.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Studying these topics provides learners with a robust foundation in network security, including the understanding of protocols, attack recognition, and security controls.
- GS2.** Learners gain practical skills in configuring and securing network components, effectively mitigating attacks, and utilizing critical security tools and technologies.
- GS3.** Learners develop expertise in the management of security operations, intrusion detection, and the use of SIEM systems, making them well-prepared for roles in network security and cybersecurity operations.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Network Security and L2/L3 Attacks</i>	9	2	-	-
PC1. Understand the concept of Network Security and Network Security Protocols.	-	-	-	-
PC2. Understand the Attacks at Layer 3 and Layer 2	-	-	-	-
PC3. Understand the CAM Table and CAM Flooding Attacks.	-	-	-	-
PC4. Understand the Switch Port Security and Switch Port Violations.	-	-	-	-
PC5. Understand the concept to prevent CAM Flooding Attacks by using Switch Port Security	-	-	-	-
<i>Network Security Control</i>	9	2	1	-
PC6. Understand the concept of Network Security Controls and Network Access controls	-	-	-	-
PC7. Understand the concept of Firewalls, DMZ, VPN, IDS, IPS, and Network Traffic Monitoring and Analysis.	-	-	-	-
PC8. Understand the Working of VPN, VPN Components and VPN Types.	-	-	-	-
<i>Intrusion Detection System(IDS)</i>	8	2	1	-
PC9. Understand the working of IP Security, Protocols used in IPSec, the Security Architecture of IPSec, and Modes of IPSec.	-	-	-	-
PC10. Understand the concept of Intrusion Detection.	-	-	-	-
<i>Understanding SOC</i>	8	2	1	-
PC11. Understand SOC, NOC, and the Functions of SOC.	-	-	-	-
PC12. Understand SIEM and Security Operations and Management	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	34	8	3	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0905
NOS Name	Network and Infrastructure Security
Sector	Information Technology Sector
Sub-Sector	Networking and Cybersecurity
Occupation	Cyber Security
NSQF Level	4
Credits	1.5
Version	1.0
Next Review Date	29/03/2026

Qualification Pack

NIE/ITS/N0905: PERCEPTION MANAGEMENT

Description

It covers the fundamentals of perception and perceptual errors, introducing perception management and its practical applications across sectors. It includes basics of web development, WordPress customization, and social media analytics, focusing on data gathering, sentiment analysis, social media listening, and evaluation techniques. Additionally, it explores multimedia editing tools and delves into psychological warfare concepts, including propaganda and counter-propaganda strategies, emphasizing their relevance in digital and social media contexts.

Scope

The scope covers the following :

- The Scope offers a comprehensive foundation in perception, social media analytics, multimedia tools, and psychological warfare techniques. It equips learners with skills to manage and analyze digital content, optimize web platforms, and understand the dynamics of social media conversations. Emphasizing practical tools and ethical considerations, it prepares students for roles in digital marketing, media monitoring, data analysis, and strategic communication across diverse sectors. The course also highlights the importance of countering misinformation through propaganda and psychological warfare awareness.

Elements and Performance Criteria

Understanding Perception and Perceptual Processes

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the stages of the perceptual process and how sensory inputs are interpreted.
- PC2.** Identify common perceptual errors and their impact on information processing.
- PC3.** Describe perception management and illustrate its applications across sectors such as marketing, defense, and public relations.

Social Media Analytics and Data Management

To be competent, the user/individual on the job must be able to:

- PC4.** Demonstrate knowledge of social media data gathering tools and platforms.
- PC5.** Apply social media monitoring and listening tools to analyze online conversations and sentiment.
- PC6.** Conduct basic sentiment analysis and social network visualization using software tools.

Web Development and Snippet Customization

To be competent, the user/individual on the job must be able to:

- PC7.** Understand the basics of WordPress and its role in website creation.
- PC8.** Customize code snippets within WordPress for enhanced website functionality.
- PC9.** Optimize websites for performance, user experience, and SEO best practices.

Multimedia Editing and Psychological Warfare

To be competent, the user/individual on the job must be able to:

- PC10.** Use multimedia tools to edit images, sound, and video content effectively.

Qualification Pack

PC11. Explain the principles and objectives of psychological warfare and propaganda.

PC12. Identify and apply counter-propaganda techniques to manage information influence.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understanding perceptual processes, errors, and how perception management influences various sectors.

KU2. Grasping social media metrics, data gathering, sentiment analysis, and related legal and ethical challenges.

KU3. Basic knowledge of WordPress, snippet customization, and website optimization.

KU4. Familiarity with multimedia editing tools and concepts of psychological warfare, propaganda, and counter-propaganda.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Ability to interpret social media data, sentiment, and perceptual processes to make informed decisions.

GS2. Understanding and managing perceptions, sentiments, and opinions, essential for both digital content creation and psychological warfare awareness.

GS3. Skilled in WordPress customization, social media analytics, monitoring, listening tools, and multimedia editing software.

GS4. Competence in gathering, analyzing, and visualizing social media metrics and network data using specialized software.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understanding Perception and Perceptual Processes</i>	55	29	-	-
PC1. Explain the stages of the perceptual process and how sensory inputs are interpreted.	-	-	-	-
PC2. Identify common perceptual errors and their impact on information processing.	-	-	-	-
PC3. Describe perception management and illustrate its applications across sectors such as marketing, defense, and public relations.	-	-	-	-
<i>Social Media Analytics and Data Management</i>	-	-	-	-
PC4. Demonstrate knowledge of social media data gathering tools and platforms.	-	-	-	-
PC5. Apply social media monitoring and listening tools to analyze online conversations and sentiment.	-	-	-	-
PC6. Conduct basic sentiment analysis and social network visualization using software tools.	-	-	-	-
<i>Web Development and Snippet Customization</i>	-	-	-	-
PC7. Understand the basics of WordPress and its role in website creation.	-	-	-	-
PC8. Customize code snippets within WordPress for enhanced website functionality.	-	-	-	-
PC9. Optimize websites for performance, user experience, and SEO best practices.	-	-	-	-
<i>Multimedia Editing and Psychological Warfare</i>	-	-	-	-
PC10. Use multimedia tools to edit images, sound, and video content effectively.	-	-	-	-
PC11. Explain the principles and objectives of psychological warfare and propaganda.	-	-	-	-
PC12. Identify and apply counter-propaganda techniques to manage information influence.	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	55	29	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0905
NOS Name	PERCEPTION MANAGEMENT
Sector	Information Technology Sector
Sub-Sector	
Occupation	CYBER SECURITY
NSQF Level	5
Credits	4.2
Version	1.0
Next Review Date	NA

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. The question papers for the theory and practical exams are set by the Examination wing (assessor) of NIELIT HQS.
2. The assessor assigns roll number
3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.
4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.
5. Pass percentage would be 50% marks in each component.

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6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).
7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.
8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.
9. The examination will be conducted in English language only.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ITS/N0907.Operating System Basics-	9	4	0	0	13	21
NIE/ITS/N0908.Computer Network-	55	27	0	0	82	30
NIE/ITS/N0906.ARTIFICIAL INTELLIGENCE / MACHINE LEARNING AND DISRUPTIVE TECHNOLOGIES	17	9	0	0	26	25
NIE/ITS/N0905.PERCEPTION MANAGEMENT	55	29	0	0	84	15
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	9
Total	156	99	0	0	255	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.